

The Hopfield Model (2)

BCSC 10th AI Study Session #3

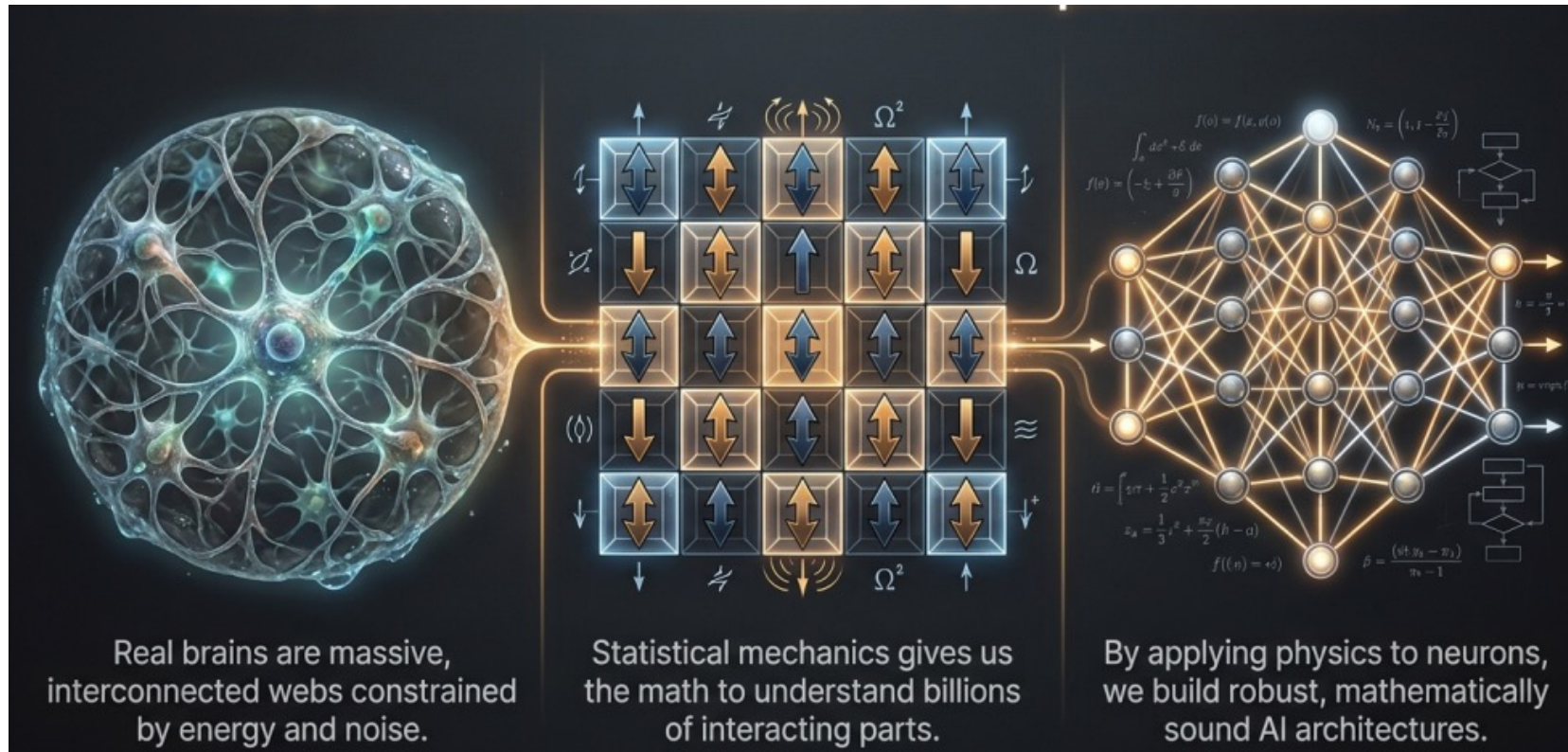
Seoyoon Jeong

Introduction

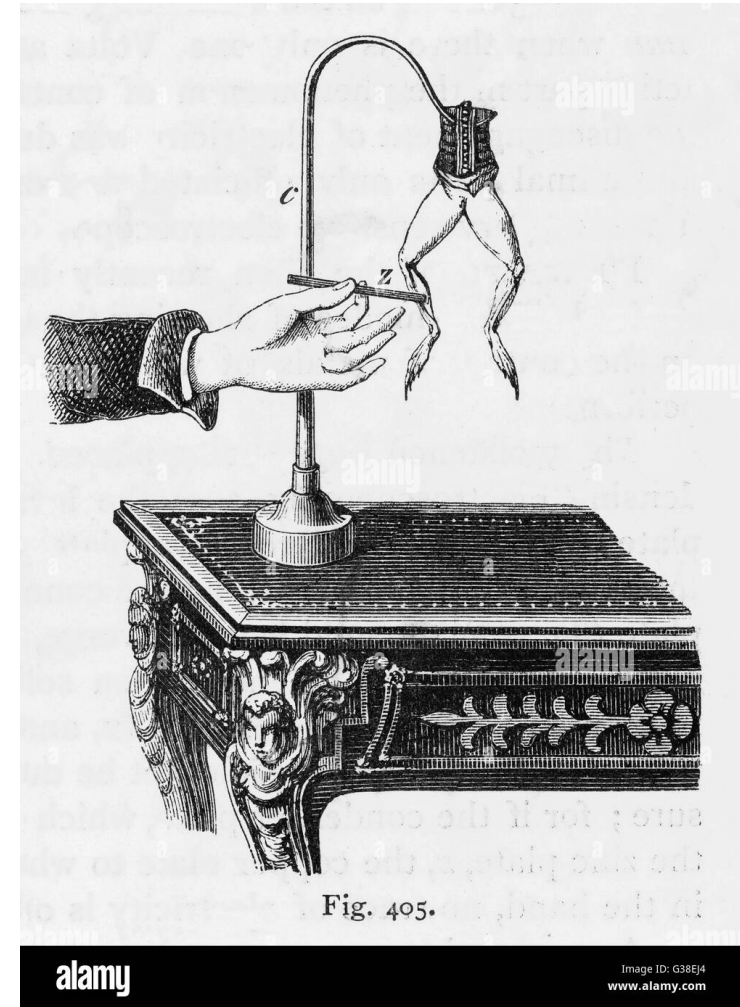
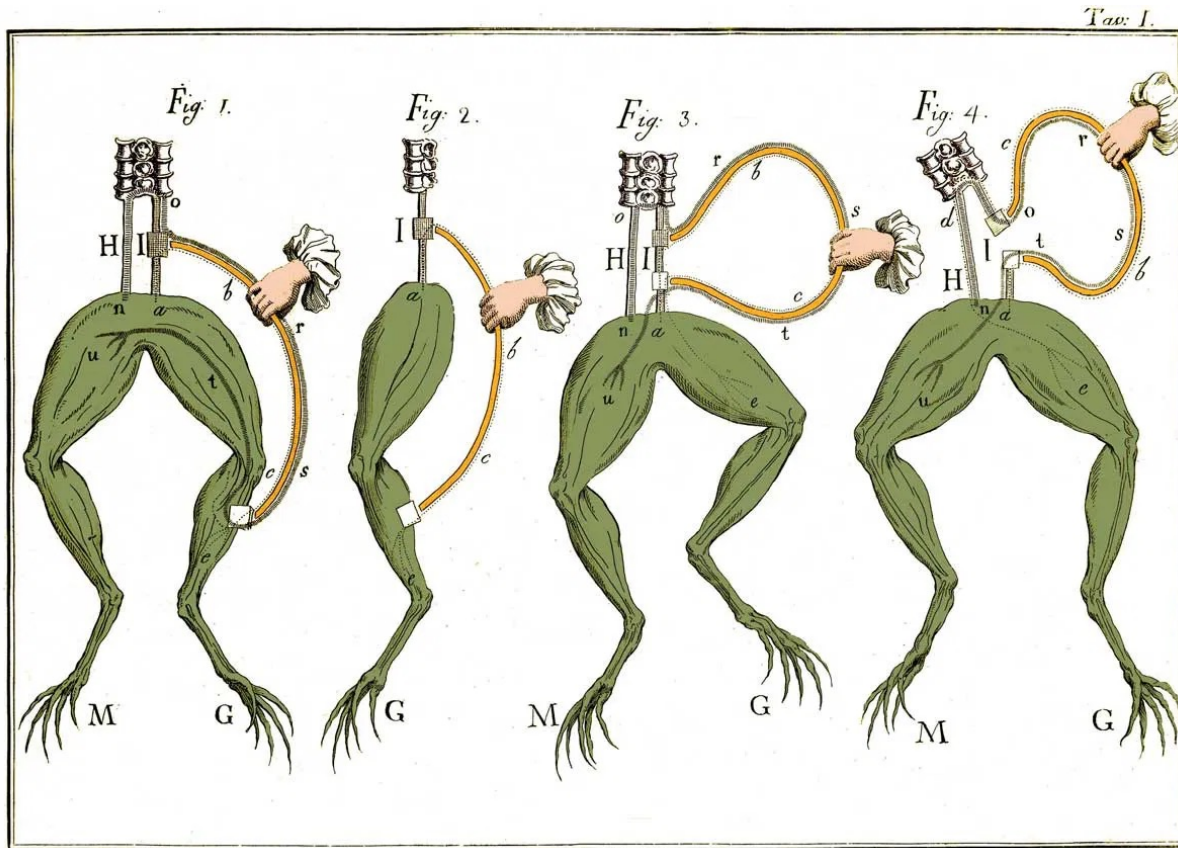
- 3 Keywords for today
 - Magnetic Analogy
 - Noise
 - 13.8%



Introduction



Introduction



Introduction



Hermann von Helmholtz

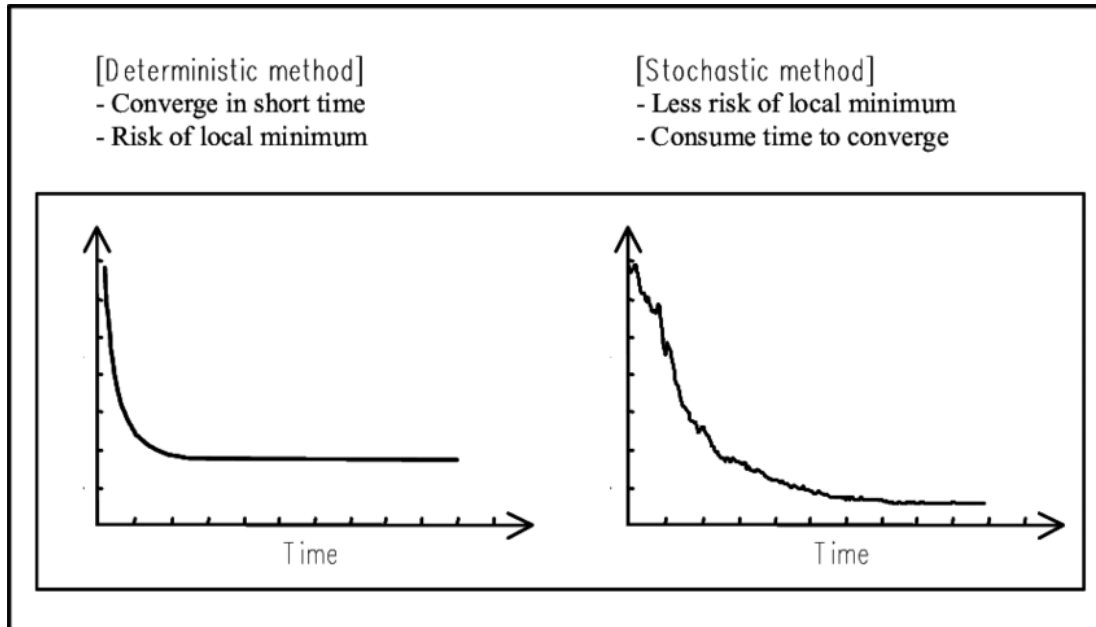


I think the facts leave no doubt that the very mightiest among the chemical forces are of electric origin. The atoms cling to their electric charges, and opposite electric charges cling to each other.

AZ QUOTES

Statistical Mechanics of Magnetic Systems

- Stochastic vs. Deterministic



Effects on
Human Body

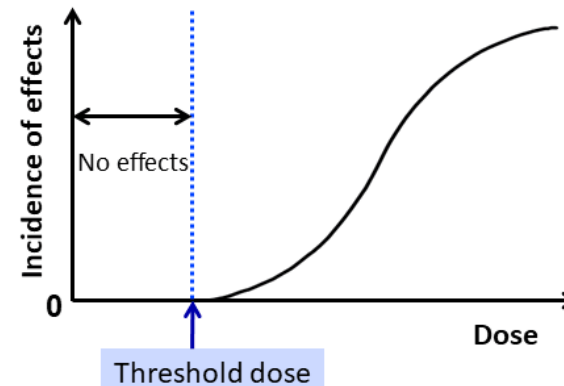
Deterministic Effects and Stochastic Effects

Deterministic effects

(Hair loss, cataract, skin injury, etc.)

When a number of people were exposed to the same dose of radiation and certain symptoms appear in 1% of them, said dose is considered to be the threshold dose.

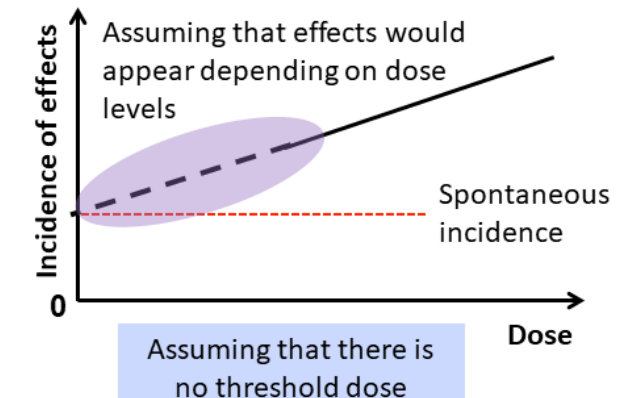
(2007 Recommendations of the International Commission on Radiological Protection (ICRP))



Stochastic effects

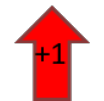
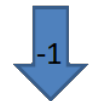
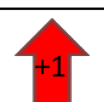
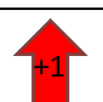
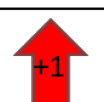
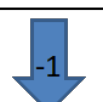
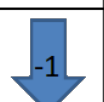
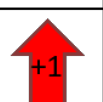
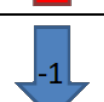
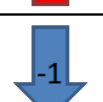
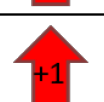
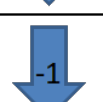
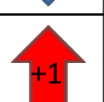
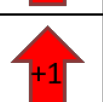
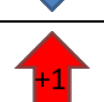
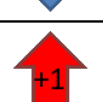
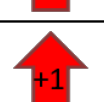
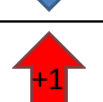
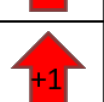
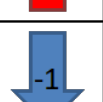
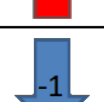
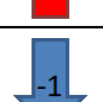
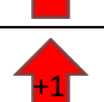
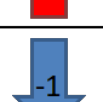
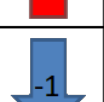
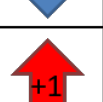
(Cancer, leukemia, hereditary effects, etc.)

Effects of radiation exposure under certain doses are not clear because effects of other cancer-promoting factors such as smoking and drinking habits are too large. However, the ICRP specifies the standards for radiological protection for such low-dose exposures, assuming that they may have some effects as well.



Statistical Mechanics of Magnetic Systems

- Ising Model
 - In a cold, deterministic system, a neuron simply follows the crowd.

$$h_i = \sum_j w_{ij} S_j + h^{ext}$$

2.34. Local Magnetic Field

$$H = -\frac{1}{2} \sum_{ij} w_{ij} S_i S_j - h^{ext} \sum_i S_i$$

2.35. Energy Function / Hamiltonian

Statistical Mechanics of Magnetic Systems

- Finite Temperature Dynamics
 - Thermal fluctuations (If the temperature is not very low)
- Glauber dynamics
 - $g(h) = f_\beta(h) \equiv \frac{1}{1+\exp(-2\beta h)}$
 - $\beta = \frac{1}{k_B T}$ where k_B is Boltzmann's constant with value 1.38×10^{-16} erg/K

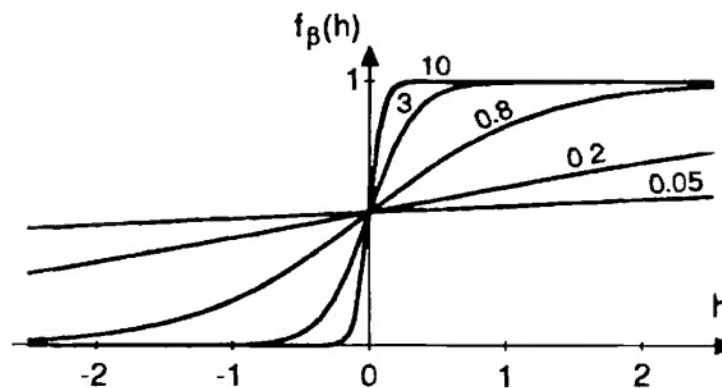


FIGURE 2.8 The sigmoid function $f_\beta(h)$ given by (2.37) for several values of β . This is sometimes called the *logistic* function.

Statistical Mechanics of Magnetic Systems

- A single spin in Equilibrium

- Average magnetization $\langle S \rangle \equiv \text{Prob}(+1) \cdot (+1) + \text{Prob}(-1) \cdot (-1)$
$$= \frac{1}{1+e^{-2\beta h}} - \frac{1}{1+e^{2\beta h}} = \frac{e^{\beta h}}{e^{\beta h}+e^{-\beta h}} - \frac{e^{-\beta h}}{e^{-\beta h}+e^{\beta h}}$$
$$= \tanh(\beta h)$$

$$\tanh(x) \equiv \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

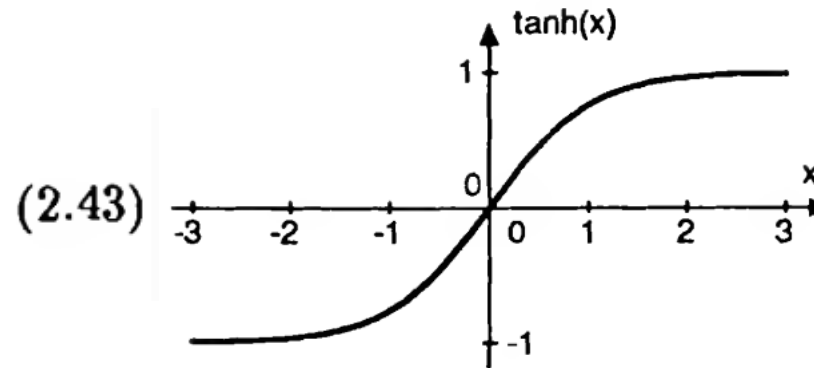


FIGURE 2.9 The function $\tanh(x)$ defined by equation (2.43).

Statistical Mechanics of Magnetic Systems

- Ferromagnet
 - Phase transition

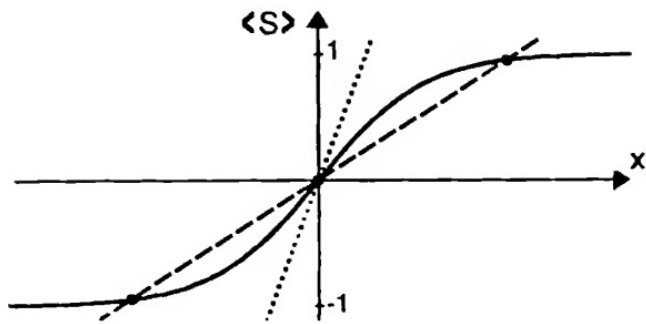


FIGURE 2.12 Solutions to equation (2.47) are intersections between the straight line $\langle S \rangle = x/\beta J$ and the sigmoid curve $\langle S \rangle = \tanh(x)$. When $\beta J \leq 1$ (dotted line) there is only one solution, at $\langle S \rangle = 0$, whereas for $\beta J > 1$ (dashed line) there are three solutions.

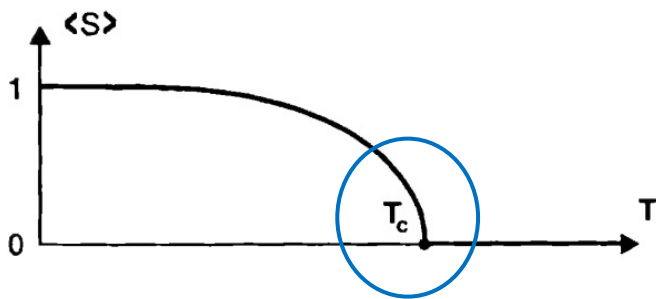


FIGURE 2.13 The positive solution of equation (2.47) as a function of temperature.

Ferromagnetic ↑ ↑ ↑ ↑ ↑ ↑ ↑	Below T_c , spins are aligned parallel in magnetic domains
Antiferromagnetic ↑ ↓ ↑ ↓ ↑ ↓ ↑	Below T_N , spins are aligned antiparallel in magnetic domains
Ferrimagnetic ↑ ↓ ↑ ↓ ↑ ↓ ↑	Below T_c , spins are aligned antiparallel but do not cancel
Paramagnetic ↑ ↓ ↑ ↑ ↓ ↑ ↓	Spins are randomly oriented (any of the others above T_c or T_N)

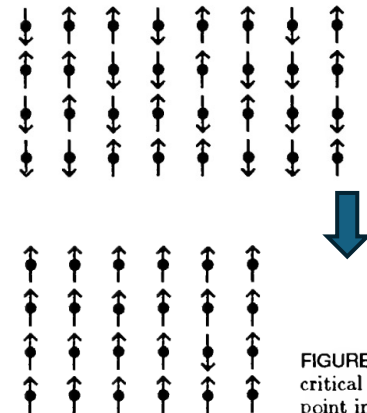


FIGURE 2.7 A very simplified picture of a magnetic material described by an Ising model.

FIGURE 2.11 An Ising ferromagnet well below the critical temperature T_c . Almost all of the spins point in the same direction.

Statistical Mechanics of Magnetic Systems

- Mean Field Theory
 - Average background field

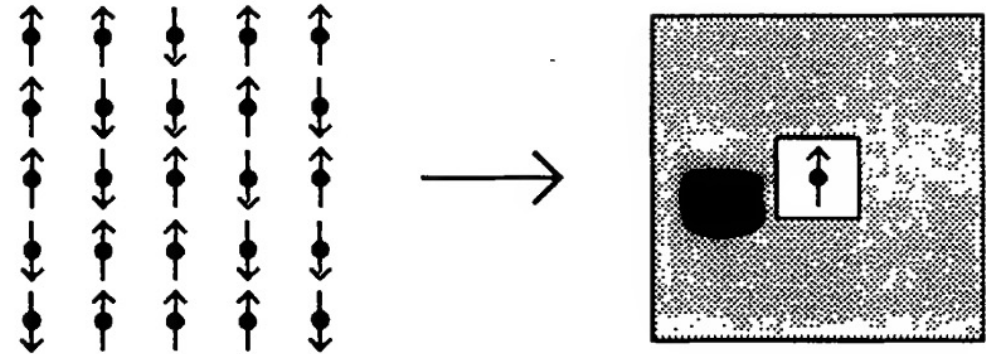
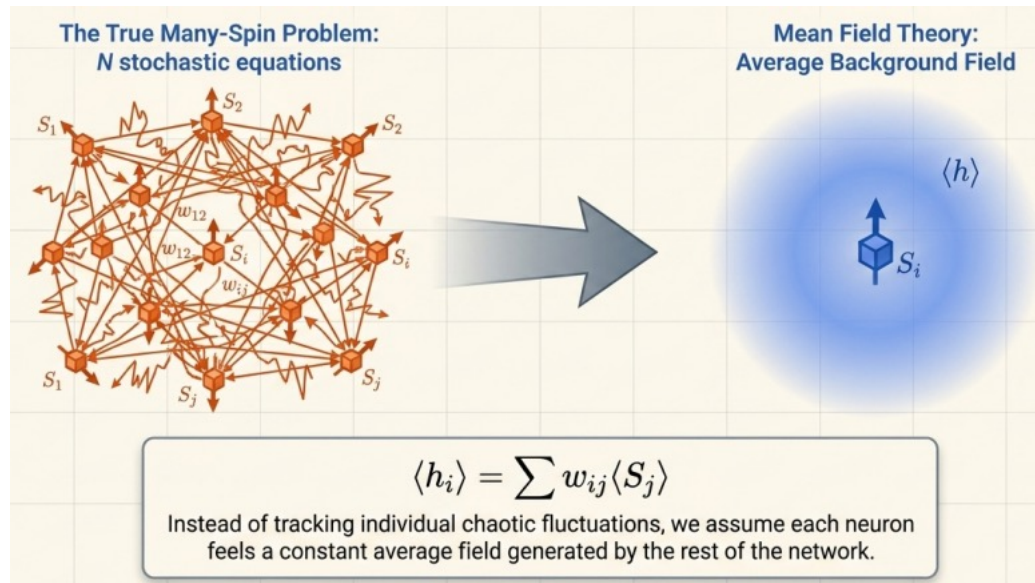
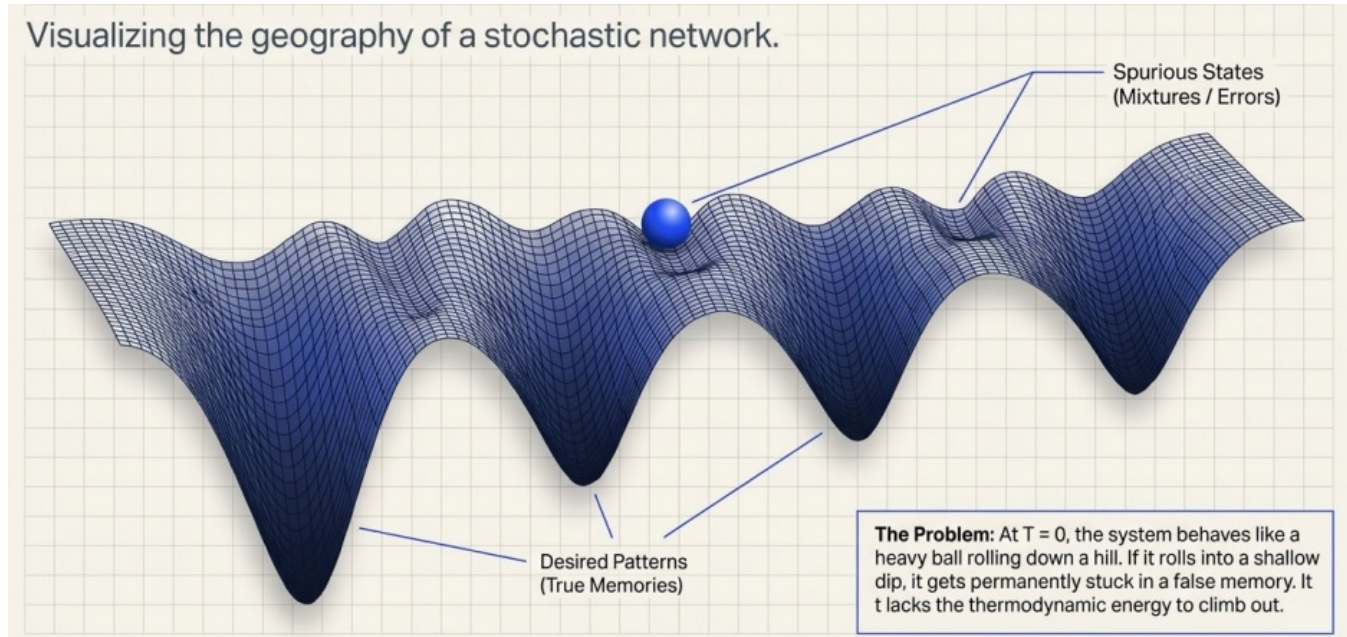


FIGURE 2.10 Mean field theory. All the spins but one are replaced by their average values.

Stochastic Networks

- Spurious local minima



- Thermal fluctuations

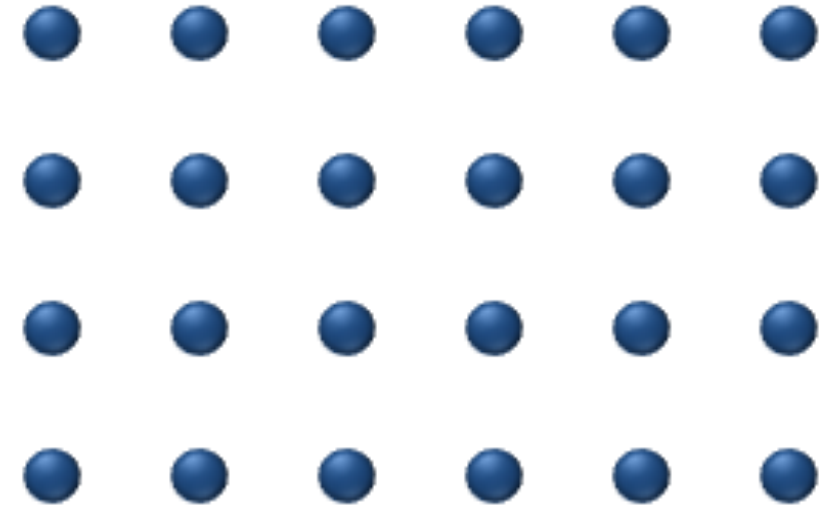


Image Source: Wikipedia

Stochastic Networks

- Pseudo-temperature T

- By setting $k_B = 1$ for simplicity, $\beta \equiv \frac{1}{T}$



- Glauber dynamics

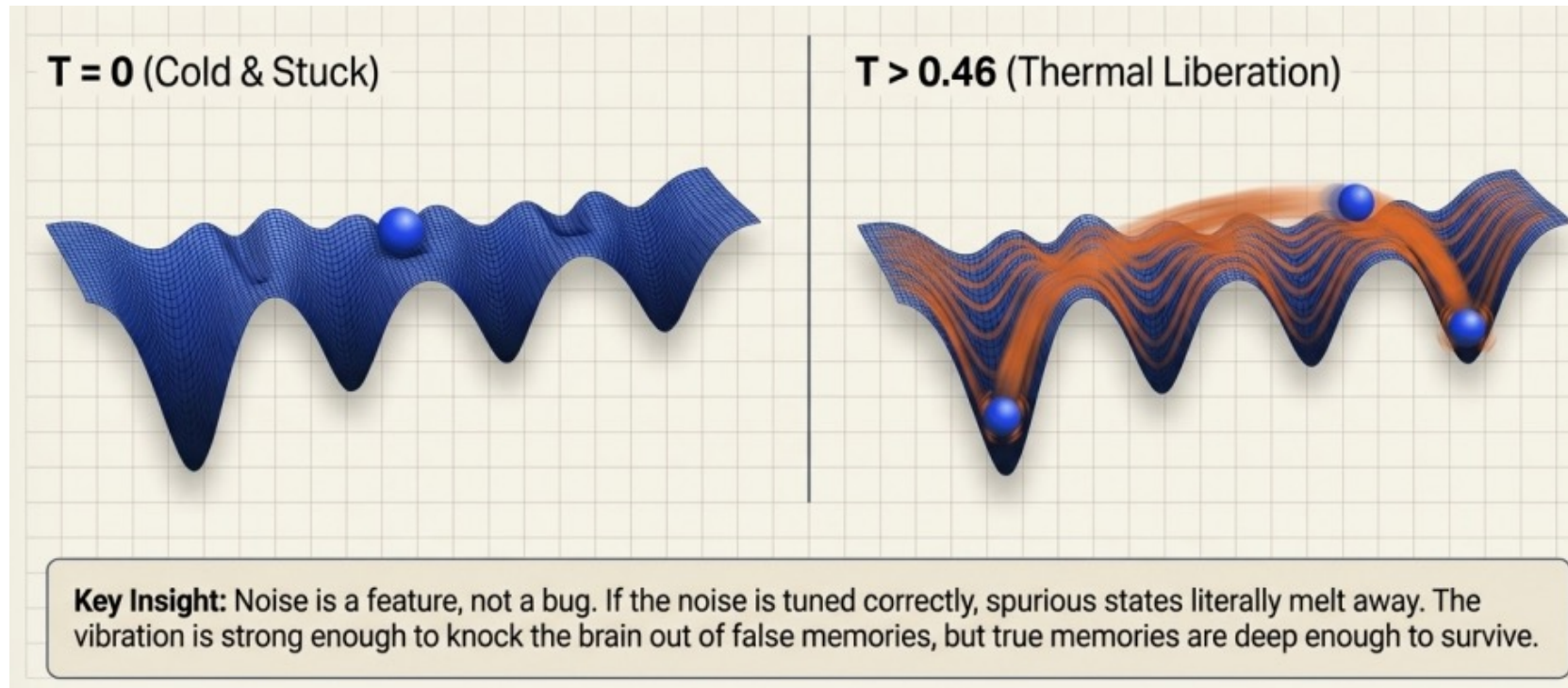
- $g(h) = f_\beta(h) \equiv \frac{1}{1 + \exp(-2\beta h)}$

- $\beta = \frac{1}{k_B T}$ where k_B is Boltzmann's constant with value 1.38×10^{-16} erg/K

- Refer to T as the temperature even though pseudo-temperature

Stochastic Networks

- Why noise is actually good



Stochastic Networks

- Why noise is actually good

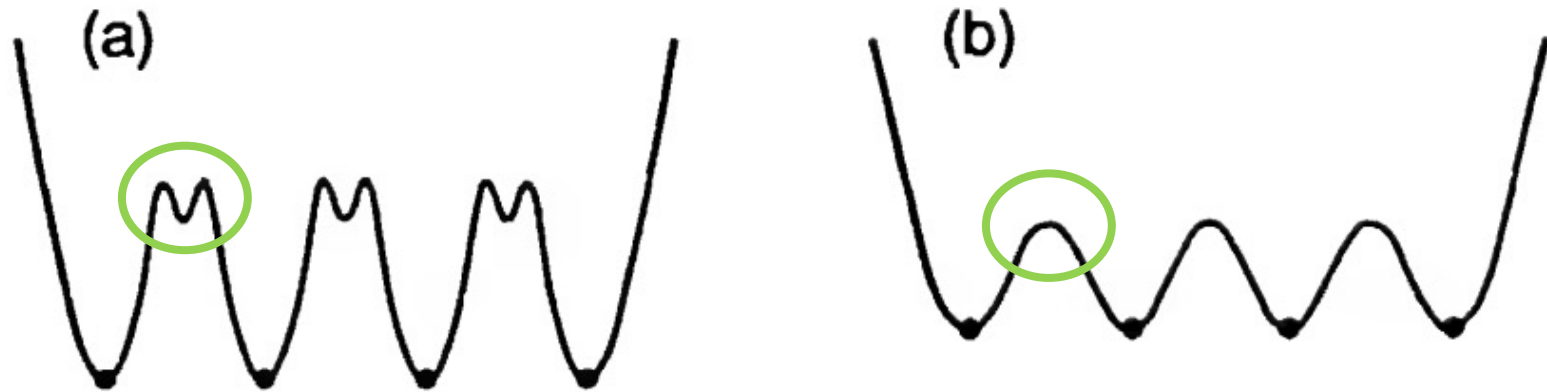


FIGURE 2.15 Schematic illustration of energy landscapes for $p \ll N$. (a) One can think of the mixture states as small dips between the desired pattern valleys. (b) At high enough temperature there are no mixture states.

A Curious Tangent: Optimization

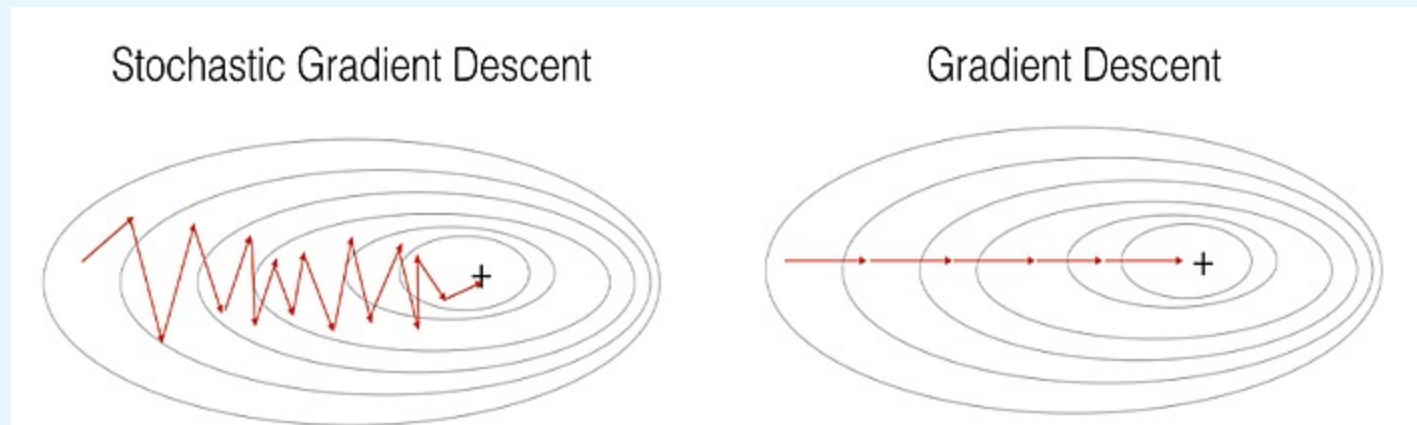
- Vanilla SGD

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

θ : Weight parameter

η : Learning Rate

$\nabla L(\theta_t)$: Gradient of the Loss function at θ_t



A Curious Tangent: Optimization

- SGD with Momentum

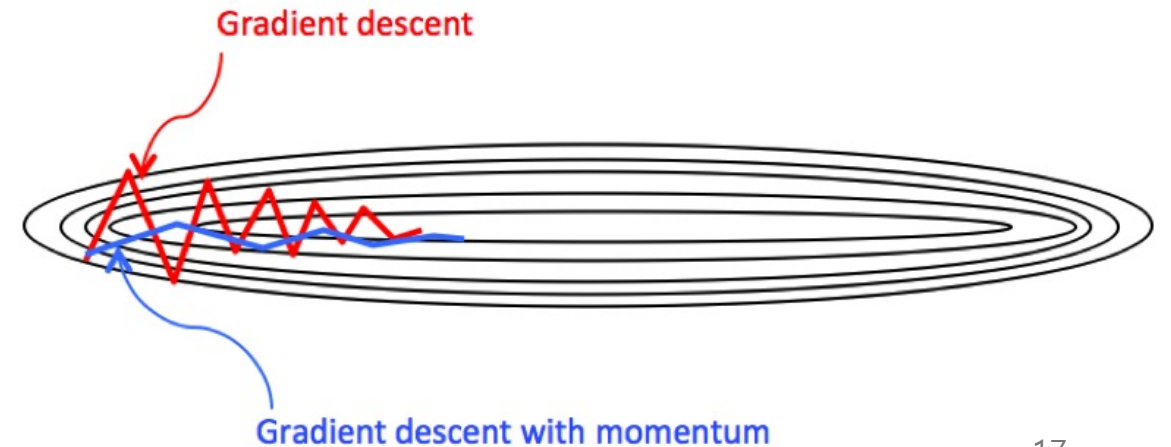
$$\begin{aligned}\theta_{t+1} &= \theta_t - v_t \\ v_t &= \gamma v_{t-1} + \eta \nabla L(\theta_t)\end{aligned}$$

θ : weight parameter

η : Learning Rate

$\nabla L(\theta_t)$: Gradient of the Loss function at θ_t

v_t : velocity



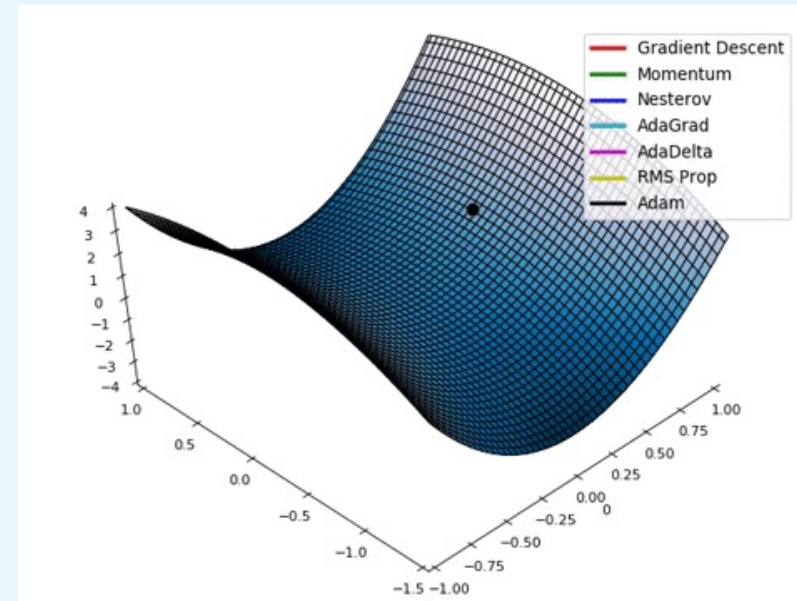
A Curious Tangent: Optimization

- Adam: Adaptive Moment Estimation

```
input :  $\gamma$  (lr),  $\beta_1, \beta_2$  (betas),  $\theta_0$  (params),  $f(\theta)$  (objective)
         $\lambda$  (weight decay), amsgrad, maximize,  $\epsilon$  (epsilon)
initialize :  $m_0 \leftarrow 0$  ( first moment),  $v_0 \leftarrow 0$  (second moment)

for  $t = 1$  to ... do
    if maximize :
         $g_t \leftarrow -\nabla_{\theta} f_t(\theta_{t-1})$ 
    else
         $g_t \leftarrow \nabla_{\theta} f_t(\theta_{t-1})$ 
    if  $\lambda \neq 0$ 
         $g_t \leftarrow g_t + \lambda \theta_{t-1}$ 
     $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$ 
     $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$ 
     $\widehat{m}_t \leftarrow m_t / (1 - \beta_1^t)$ 
    if amsgrad
         $v_t^{max} \leftarrow \max(v_{t-1}^{max}, v_t)$ 
         $\widehat{v}_t \leftarrow v_t^{max} / (1 - \beta_2^t)$ 
    else
         $\widehat{v}_t \leftarrow v_t / (1 - \beta_2^t)$ 
     $\theta_t \leftarrow \theta_{t-1} - \gamma \widehat{m}_t / (\sqrt{\widehat{v}_t} + \epsilon)$ 

return  $\theta_t$ 
```



A Curious Tangent: Optimization

- AdamW: Adam with Weight decay

```
input :  $\gamma(\text{lr}), \beta_1, \beta_2(\text{betas}), \theta_0(\text{params}), f(\theta)(\text{objective}), \epsilon (\text{weight decay}), \text{amsgrad}, \text{maximize}$   
initialize :  $m_0 \leftarrow 0$  (first moment),  $v_0 \leftarrow 0$  (second moment)
```

```
for  $t = 1$  to ... do
```

```
  if maximize :
```

```
     $g_t \leftarrow -\nabla_{\theta} f_t(\theta_{t-1})$ 
```

```
  else
```

```
     $g_t \leftarrow \nabla_{\theta} f_t(\theta_{t-1})$ 
```

```
     $\theta_t \leftarrow \theta_{t-1} - \gamma \lambda \theta_{t-1}$ 
```

```
     $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$ 
```

```
     $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$ 
```

```
     $\widehat{m}_t \leftarrow m_t / (1 - \beta_1^t)$ 
```

```
    if amsgrad
```

```
       $v_t^{max} \leftarrow \max(v_{t-1}^{max}, v_t)$ 
```

```
       $\widehat{v}_t \leftarrow v_t^{max} / (1 - \beta_2^t)$ 
```

```
    else
```

```
       $\widehat{v}_t \leftarrow v_t / (1 - \beta_2^t)$ 
```

```
     $\theta_t \leftarrow \theta_t - \gamma \widehat{m}_t / (\sqrt{\widehat{v}_t} + \epsilon)$ 
```

```
return  $\theta_t$ 
```

if $\lambda \neq 0$

$g_t \leftarrow g_t + \lambda \theta_{t-1}$

[Submitted on 14 Nov 2017 (v1), last revised 4 Jan 2019 (this version, v3)]

Decoupled Weight Decay Regularization

Ilya Loshchilov, Frank Hutter

[View PDF](#)

L_2 regularization and weight decay regularization are equivalent for standard stochastic gradient descent (when rescaled by the learning rate), but as we demonstrate this is *not* the case for adaptive gradient algorithms, such as Adam. While common implementations of these algorithms employ L_2 regularization (often calling it "weight decay" in what may be misleading due to the inequivalence we expose), we propose a simple modification to recover the original formulation of weight decay regularization by *decoupling* the weight decay from the optimization steps taken w.r.t. the loss function. We provide empirical evidence that our proposed modification (i) decouples the optimal choice of weight decay factor from the setting of the learning rate for both standard SGD and Adam and (ii) substantially improves Adam's generalization performance, allowing it to compete with SGD with momentum on image classification datasets (on which it was previously typically outperformed by the latter). Our proposed decoupled weight decay has already been adopted by many researchers, and the community has implemented it in TensorFlow and PyTorch; the complete source code for our experiments is available at [this https URL](https://github.com/Loshchilov/Decoupled-Weight-Decay-Regularization)

Comments: Published as a conference paper at ICLR 2019

Subjects: **Machine Learning (cs.LG)**; Neural and Evolutionary Computing (cs.NE); Optimization and Control (math.OC)

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(or [arXiv:1711.05101v3](https://arxiv.org/abs/1711.05101v3) [cs.LG] for this version)

<https://doi.org/10.48550/arXiv.1711.05101>

Capacity of the Stochastic Network

- Load parameter $\alpha \equiv \frac{p}{N}$
 - p = Number of stored patterns (memories)
 - N = Number of units (neurons)

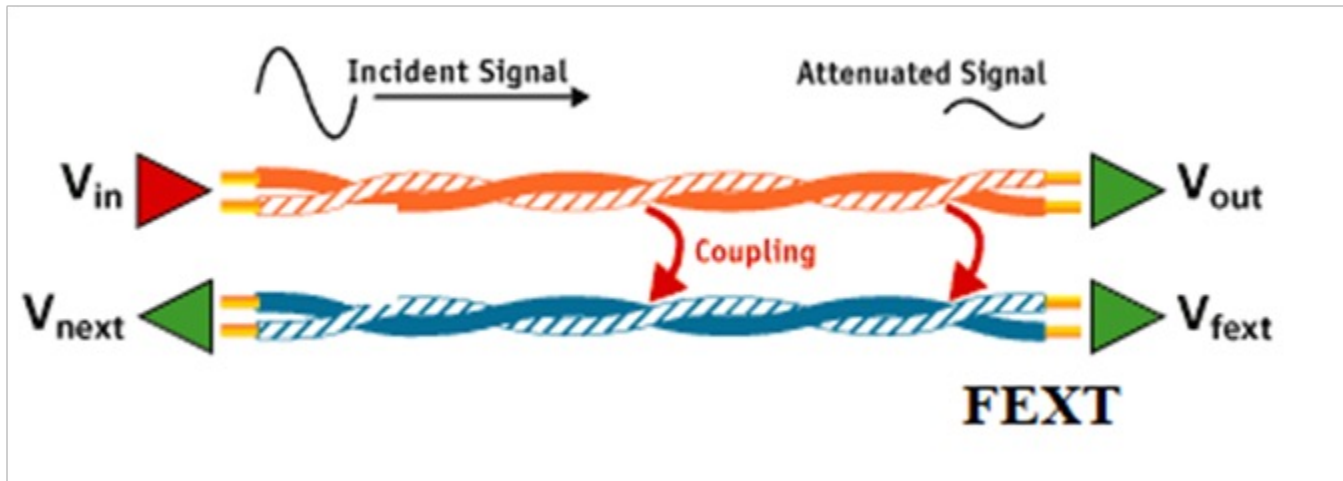
$$N = 1000, \quad p = 50$$

$$\alpha = \frac{50}{1000} = 0.05$$

$$N = 1000, \quad p = 300$$

$$\alpha = 0.3$$

- Crosstalk



Capacity of the Stochastic Network

- Overlap $m_\nu = \frac{1}{N} \sum_i \xi_i^\nu \langle S_i \rangle$
 - Successful retrieval if $m_\nu \approx 1$
 - Retrieval failure if $m_\nu \approx 0$
- Crosstalk

$$r = \frac{1}{\alpha} \sum_{\nu \neq 1} m_\nu^2, \quad (2.59)$$

$$m_\nu = \frac{1}{N} \sum_i \xi_i^\nu \tanh\left(\beta \sum_\mu \xi_i^\mu m_\mu\right) \quad (2.60)$$

and, for $\nu \neq 1$, separate out explicitly the terms with $\mu = 1$ and with $\mu = \nu$:

$$m_\nu = \frac{1}{N} \sum_i \xi_i^\nu \xi_i^1 \tanh\left[\beta\left(m_1 + \xi_i^\nu \xi_i^1 m_\nu + \sum_{\mu \neq 1, \nu} \xi_i^\mu \xi_i^1 m_\mu\right)\right]. \quad (2.61)$$

$$\langle S_i \rangle = \tanh\left(\beta \sum_\mu \xi_i^\mu m_\mu\right)$$

Capacity of the Stochastic Network

The first term in the argument of the tanh is large (order 1) by hypothesis, because it is pattern 1 that we are retrieving, and the last term is large because there are about p terms in it. But the second term is small, order $1/\sqrt{N}$, so we can expand:

$$m_\nu = \frac{1}{N} \sum_i \xi_i^\nu \xi_i^1 \tanh \left[\beta \left(m_1 + \sum_{\mu \neq 1, \nu} \xi_i^\mu \xi_i^1 m_\mu \right) \right] + \frac{\beta}{N} \sum_i \left\{ 1 - \tanh^2 \left[\beta \left(m_1 + \sum_{\mu \neq 1, \nu} \xi_i^\mu \xi_i^1 m_\mu \right) \right] \right\} m_\nu \quad (2.62)$$

using $\frac{d}{dx} \tanh(x) = 1 - \tanh^2(x)$. We now *assume* that the small overlaps m_μ , $\mu \neq 1$, are independent random variables with mean zero and variance $\alpha r/p$, as suggested by (2.59). In the second line of (2.62), $\xi_i^\mu \xi_i^1$ is random and independent of m_μ , so by the central limit theorem the site average $N^{-1} \sum_i$ is effectively an average over a Gaussian “noise” $\sum_{\mu \neq 1, \nu} \xi_i^\mu \xi_i^1 m_\mu$ of variance αr . Thus (2.62) reduces to

$$m_\nu = \frac{1}{N} \sum_i \xi_i^\nu \xi_i^1 \tanh \left[\beta \left(m_1 + \sum_{\mu \neq 1, \nu} \xi_i^\mu \xi_i^1 m_\mu \right) \right] + \beta m_\nu - \beta q m_\nu \quad (2.63)$$

or

$$m_\nu = \frac{N^{-1} \sum_i \xi_i^\nu \xi_i^1 \tanh[\beta(m_1 + \sum_{\mu \neq 1, \nu} \xi_i^\mu \xi_i^1 m_\mu)]}{1 - \beta(1 - q)} \quad (2.64)$$

where

$$q = \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \tanh^2[\beta(m_1 + \sqrt{\alpha r} z)]. \quad (2.65)$$

Now we can calculate r . We just square (2.64)

$$m_\nu^2 = \left[\frac{1}{1 - \beta(1 - q)} \right]^2 \frac{1}{N^2} \sum_{ij} \xi_i^\nu \xi_i^1 \xi_j^\nu \xi_j^1 \times \tanh \left[\beta \left(m_1 + \sum_{\mu \neq 1, \nu} \xi_i^\mu \xi_i^1 m_\mu \right) \right] \times \tanh \left[\beta \left(m_1 + \sum_{\mu \neq 1, \nu} \xi_j^\mu \xi_j^1 m_\mu \right) \right] \quad (2.66)$$

and average the result over patterns. Since pattern ν does not occur inside the tanh's, the pattern factors $\xi_i^\nu \xi_j^\nu$ outside the tanh's can be averaged separately, and only the $i = j$ term survives. Then the remaining average of the tanh's just gives a factor of q as in (2.62). The result is independent of ν , so from (2.59):

$$r = \frac{q}{[1 - \beta(1 - q)]^2}. \quad (2.67)$$

We also need an equation for m_1 . Using the same approach, starting again from (2.60) with $\nu = 1$, it is easy to obtain

$$m_1 = \int \frac{dz}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \tanh[\beta(m_1 + \sqrt{\alpha r} z)]. \quad (2.68)$$

The three equations (2.65), (2.67), and (2.68) can now be solved simultaneously for m_1 , q , and r . In general this must be done numerically. We examine in detail only the $T \rightarrow 0$ (or $\beta \rightarrow \infty$) limit. In this limit it is clear from (2.65) that $q \rightarrow 1$, but the quantity $C \equiv \beta(1 - q)$ remains finite. The $T \rightarrow 0$ limit lets us evaluate the integrals in (2.65) and (2.68), using the identities

$$\begin{aligned} \int \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} (1 - \tanh^2 \beta[az + b]) &\simeq \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \Big|_{\tanh^2 \beta[az+b]=0} \times \int dz (1 - \tanh^2 \beta[az + b]) \\ &= \frac{1}{\sqrt{2\pi}} e^{-b^2/2a^2} \frac{1}{a\beta} \int dz \frac{\partial}{\partial z} \tanh \beta[az + b] \\ &= \sqrt{\frac{2}{\pi}} \frac{1}{a\beta} e^{-b^2/2a^2} \end{aligned} \quad (2.69)$$

and

$$\begin{aligned} \int \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} \tanh \beta[az + b] &\xrightarrow{T \rightarrow 0} \int \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} \operatorname{sgn}[az + b] \\ &= 2 \int_{-b/a}^{\infty} \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} - 1 \\ &= \operatorname{erf}\left(\frac{b}{\sqrt{2}a}\right) \end{aligned} \quad (2.70)$$

where the error function $\operatorname{erf}(x)$ was defined in (2.16). Our three equations thus become:

$$C \equiv \beta(1 - q) = \sqrt{\frac{2}{\pi \alpha r}} \exp\left(-\frac{m^2}{2\alpha r}\right) \quad (2.71)$$

$$r = \frac{1}{(1 - C)^2} \quad (2.72)$$

$$m = \operatorname{erf}\left(\frac{m}{\sqrt{2\alpha r}}\right) \quad (2.73)$$

where we have written m for m_1 . We can find the capacity of the network by solving these three equations. Setting $y = m/\sqrt{2\alpha r}$, we obtain the equation

$$y \left(\sqrt{2\alpha} + \frac{2}{\sqrt{\pi}} e^{-y^2} \right) = \operatorname{erf}(y) \quad (2.74)$$

which is easily solved graphically as shown in Fig. 2.16.

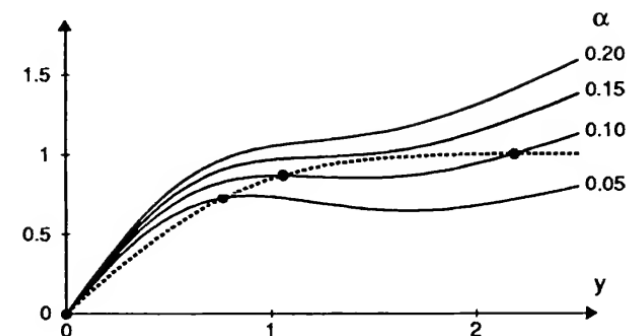


FIGURE 2.16 Graphical solution of equation (2.74). The solid lines show the left-hand side for several values of α , while the dotted line shows the right-hand side, $\operatorname{erf}(y)$. Nontrivial solutions with $m > 0$ are given by intersections away from the origin.

Capacity of the Stochastic Network

- Phase transition & Spin glass

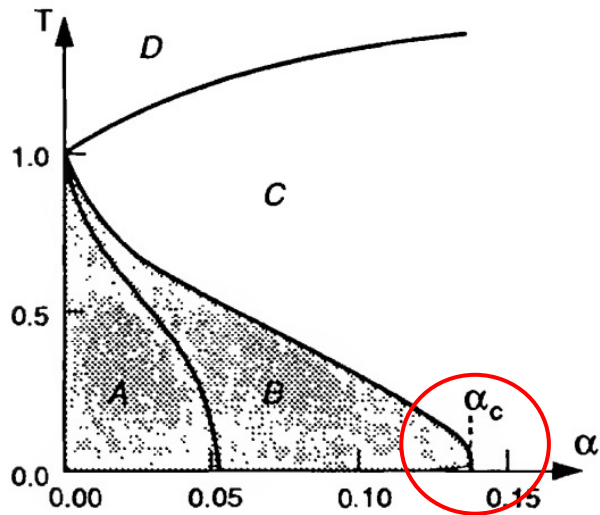


FIGURE 2.17 The phase diagram obtained by Amit et al.. The desired memory states are only stable in the shaded region.

The Phase Diagram of the Hopfield Model

By solving (2.74) we can see that there is a critical value α_c of α where the nontrivial solutions ($m \neq 0$) disappear. A numerical evaluation gives

$$\alpha_c \approx 0.138. \quad (2.75)$$

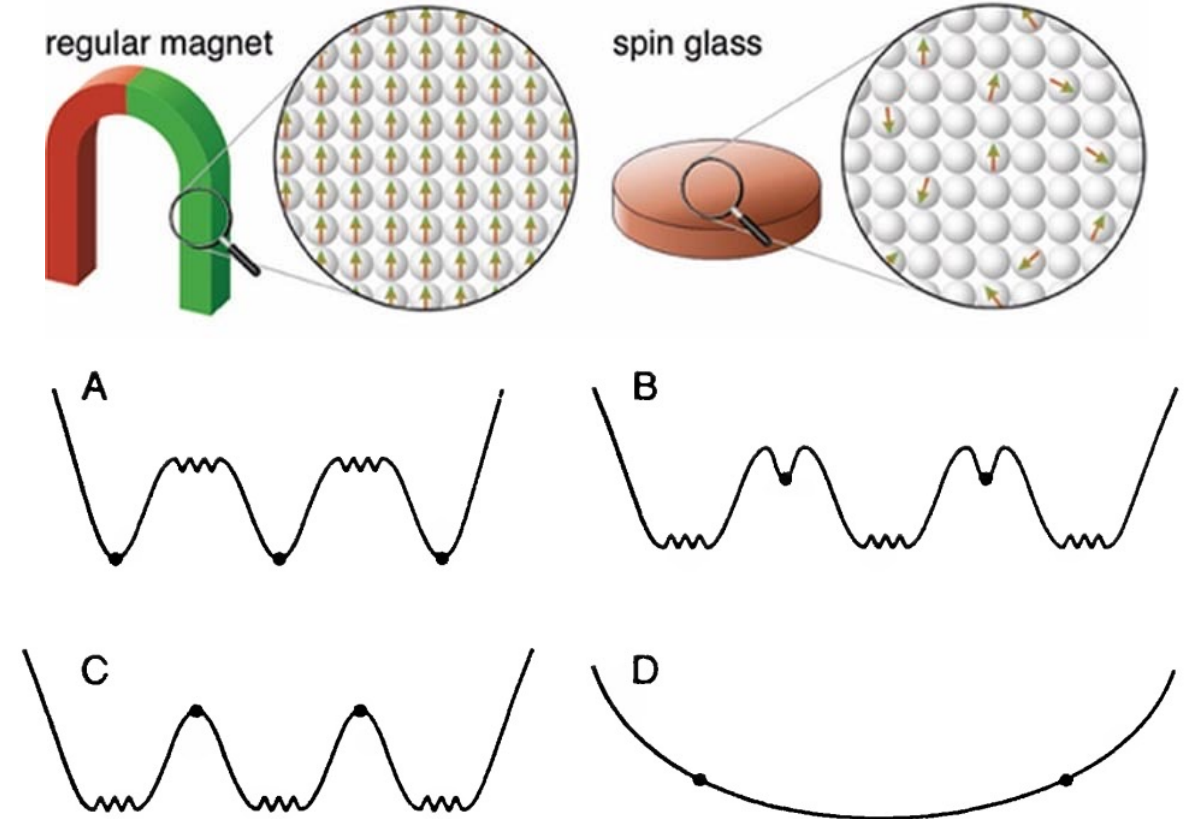


FIGURE 2.18 An attempt to visualize the energy landscape in different parts of the phase diagram. The dots show the desired memory states, while the small ripples represent spurious states. The four cases correspond to the four regions A-D of the phase diagram (figure 2.17).

Further empirical topics

- Hippocampus CA3
 - Wills et al., 2005
 - Attractor Dynamics in the Hippocampal Representation of the Local Environment

